ANALYSIS REPORT

Track 2 Official Model Verification Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Verdict

This official closure report accepts the completed `Track 2H-L` latent-state hysteresis refresh into the canonical `Track 2` evidence package.

Decision

- `track2h_latent_state_hysteresis_registry` is closed as a verified exploratory baseline.
- No `Track 2H-L` candidate is promoted over the accepted direction-parallel leaders.
- The strongest refreshed `Track 2H-L` aggregate candidate is `track2h_l_causal_tcn_offset_residual_global` .
- The accepted direction-parallel leaders remain `rcim_retuned_GBM19_Fw` , `periodic_gru_sequence_Bw` , and the accepted global neural `periodic_gru_sequence_global` .
- The closure keeps latent-state / hysteresis-aware modeling as integration evidence for later multi-head work, not as the next standalone promoted family.

Source Package

This official report consolidates these refreshed artifacts:

- metric matrix: `doc/reports/analysis/track2/Track 2 Directional Model Comparison.md` ;
- matrix summary: `output/validation_checks/track2_reference_comparison/2026-06-18-15-20-21__track2_full_directional_family_matrix_track2h_latent_state_hysteresis_track2_refresh_2026_06_18/validation_summary.yaml` ;
- per-condition metrics: `output/validation_checks/track2_reference_comparison/2026-06-18-15-20-21__track2_full_directional_family_matrix_track2h_latent_state_hysteresis_track2_refresh_2026_06_18/per_condition_metrics.csv` ;
- best-model collage report: `doc/reports/analysis/track2/best_model_collage_report/[2026-06-18]/track2_best_model_collage_report.md` ;
- multi-model curve comparison report: `doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-06-18]/track2_multi_model_curve_comparison_report.md` ;
- operator launch logs: `output/validation_checks/track2_operator_launch_logs/2026-06-18-15-20-09_track2h_latent_state_hysteresis_track2_refresh_2026_06_18` .

Candidate Refresh

The refresh added `6` candidates from `track2h_latent_state_hysteresis_registry` into the official `165` -candidate matrix.

Surface	Candidate	Family
global	`track2h_l_gru_offset_residual_global`	`track2h_l_gru_offset_residual`
Fw	`track2h_l_gru_offset_residual_Fw`	`track2h_l_gru_offset_residual`
Bw	`track2h_l_gru_offset_residual_Bw`	`track2h_l_gru_offset_residual`
global	`track2h_l_causal_tcn_offset_residual_global`	`track2h_l_causal_tcn_offset_residual`
Fw	`track2h_l_causal_tcn_offset_residual_Fw`	`track2h_l_causal_tcn_offset_residual`
Bw	`track2h_l_causal_tcn_offset_residual_Bw`	`track2h_l_causal_tcn_offset_residual`

Refreshed Source Leaders

The table ranks the refreshed source by aggregate offline Track 2 metrics.

Surface	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
Fw	`track2h_l_causal_tcn_offset_residual_Fw`	0.003476	0.003939	7.717	13.330
Bw	`track2h_l_gru_offset_residual_Bw`	0.003542	0.003984	7.736	12.831
global	`track2h_l_causal_tcn_offset_residual_global`	0.003372	0.003827	7.398	13.454

Interpretation

- `global` : causal TCN is the best `Track 2H-L` aggregate candidate, but it remains behind the accepted global neural baseline and behind the strongest probabilistic `Track 2H` global scalar evidence.
- `Fw` : causal TCN is the best `Track 2H-L` forward candidate, but its `0.003476 deg` curve MAE is far behind `rcim_retuned_GBM19_Fw` at `0.001089 deg` .
- `Bw` : GRU is the best `Track 2H-L` backward candidate, but its `0.003542 deg` curve MAE is behind `periodic_gru_sequence_Bw` at `0.002392 deg` .

Refreshed Source Leaderboard

Rank	Surface	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
1	global	`track2h_l_causal_tcn_offset_residual_global`	0.003372	0.003827	7.398	13.454
2	Fw	`track2h_l_causal_tcn_offset_residual_Fw`	0.003476	0.003939	7.717	13.330
3	Bw	`track2h_l_gru_offset_residual_Bw`	0.003542	0.003984	7.736	12.831
4	Fw	`track2h_l_gru_offset_residual_Fw`	0.003549	0.003996	7.873	12.664
5	global	`track2h_l_gru_offset_residual_global`	0.003591	0.004024	7.896	12.986
6	Bw	`track2h_l_causal_tcn_offset_residual_Bw`	0.003624	0.004098	7.903	13.135

Current Direction Leaders

These leaders are read from the matrix direction breakdown after the refresh.

Direction	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
backward	`periodic_gru_sequence_Bw`	0.002392	0.002639	5.466	14.820
forward	`rcim_retuned_GBM19_Fw`	0.001089	0.001299	2.372	4.912

The current accepted direction-parallel baselines do not change.

Multi-Index Closure

This refresh provides raw-error and visual-evidence coverage for the `Track 2H-L` source. The companion collage and overlay reports show that the latent-state candidates follow the broad TE curve trend, but they still show visible offset and high-frequency mismatch on representative conditions. The available evidence does not justify promotion on raw error, offset / continuity behavior, harmonic / phase fidelity, robustness, or deployment readiness.

The official closure therefore treats `Track 2H-L` as useful evidence that causal history can help the `global` scalar surface, while rejecting it as a standalone replacement for the accepted forward, backward, or global Track 2 leaders.

Visual Evidence

The same launcher run regenerated the visual companion reports and verified that the refreshed source appears in the visual package.

Source	Collage	Overlay Forward	Overlay Backward
`track2h_latent_state_hysteresis_registry`	6	2	2

Closeout Decision

`Track 2H-L latent-state hysteresis refresh` is closed as a verified exploratory baseline and is not promoted.

Use the latent-state hysteresis evidence as a later integration ingredient when comparing multi-head designs that combine causal state, offset handling, uncertainty, mixture pressure, and structured harmonic residuals.